Statistics: The Complete Picture

Everything We Learned in 10 Lectures (+ a Preview of What's Next)

The Journey

Describe L1–L2 → **Estimate** L3–L8 → **Test** L9–L10 → **Model** ML preview → **Cause** ML preview

L1–L2: What does the data look like? Summaries, plots, pitfalls.

L3–L8: How do we estimate unknown parameters? Properties, MLE, MAP, CIs, bootstrap.

L9–L10: Is the pattern real? Hypothesis tests, ANOVA, A/B testing.

Preview: How do variables relate? Regression, GLMs. *(in ML course)*

Preview: Did X *cause* Y? DAGs, potential outcomes, causal inference. *(in ML course)*

L1–L2: Foundations & Descriptive Statistics

L1 — Foundations

- Population vs sample
- Parameter θ vs statistic $\hat{\theta}$
- Estimand / estimator / estimate
- i.i.d. assumption
- Loss functions: L_1, L_2, L_0
- Risk $= E[\text{Loss}]$ and ERM
- Plug-in principle

L2 — Descriptive Statistics

- Mean, median, mode
- Variance, IQR, MAD
- Histograms, boxplots, ECDF
- Skewness & kurtosis
- Anscombe's quartet
- Simpson's paradox
- Always plot your data!

Key lesson: Statistics begins with a question about a *population*, answered using a *sample*.

L3–L4: Properties of Estimators

L3 — Bias, Variance, MSE

- Bias $= E[\hat{\theta}] - \theta$
- MSE $=$ Bias^2 $+$ Variance
- Bias-variance tradeoff
- Consistency
- Sufficiency (Fisher–Neyman)
- Exponential family

L4 — Fisher Information & CR

- Score function $s(\theta)$
- Fisher info $\mathcal{I}(\theta)$
- Cramér–Rao lower bound: $\mathrm{Var}(\hat{\theta}) \geq 1/\mathcal{I}(\theta)$
- Efficiency
- Stein's paradox ($d \geq 3$)

Key lesson: An estimator has a *sampling distribution*. We judge it by bias, variance, and MSE. Fisher information tells us the best we can possibly do.

L5–L6: Point Estimation — MLE & MAP

L5 — MLE

- Method of Moments
- Likelihood $\rightarrow$ log-likelihood
- MLE: $\hat{\theta} = \arg\max \ell(\theta)$
- MLE is consistent, asymp. normal,
  efficient (achieves CR bound)
- MLE $=$ least squares (Normal)
- MLE $=$ cross-entropy (logistic)

L6 — MAP & Bayesian

- Prior $\times$ likelihood $\propto$ posterior
- Conjugate priors
- MAP: $\hat{\theta} = \arg\max$ posterior
- **Regularization $=$ MAP:**
  Gaussian prior $\rightarrow$ Ridge (L_2)
  Laplace prior $\rightarrow$ Lasso (L_1)
- MLE vs MAP vs full Bayesian

Key lesson: MLE $=$ let the data speak. MAP $=$ data $+$ prior beliefs. Regularization in ML is Bayesian estimation in disguise.

L7–L8: Sampling Distributions, CIs & Bootstrap

L7 — Sampling Distributions

- $\hat{\theta}$ is a random variable
- Monte Carlo simulation
- CLT: $\bar{X} \approx N(\mu, \sigma^2/n)$
- SD $\neq$ SE
- SE $= \sigma/\sqrt{n}$ ($\sqrt{n}$ law)
- SE via Fisher info for MLEs

L8 — CIs & Bootstrap

- Wald CI: $\hat{\theta} \pm z \cdot$ SE
- Wilson CI for proportions
- t-interval, delta method
- **Bootstrap:** resample $\rightarrow$ SE, CI
- Percentile & BCa intervals
- Two paths: analytical vs computational

Key lesson: A point estimate without uncertainty is useless.
CIs quantify "how much could the estimate wiggle?"

L9–L10: Hypothesis Testing & Classical Tests

L9 — Hypothesis Testing

- H_0 vs H_1, test statistic
- Type I / II errors
- p-value: $P(\text{data} \mid H_0)$
- Statistical vs practical significance
- Power & sample size planning
- Permutation tests
- Multiple testing (Bonferroni, BH)

L10 — Classical Tests & LRT

- LRT: $-2 \log \Lambda \to \chi^2_k$
- One-sample & paired t-test
- Two-sample t (Welch's)
- Test for proportions
- χ^2 goodness-of-fit
- χ^2 test of independence
- Nonparametric: Mann–Whitney, Wilcoxon

Key lesson: A p-value asks "how surprising is the data if H_0 is true?"
Small $p \neq$ large effect. Always report effect sizes.

L10 (cont.): ANOVA & A/B Testing

ANOVA

- Multiple t-tests $\rightarrow$ false positive explosion
- $\text{SS}_T = \text{SS}_B + \text{SS}_W$, $F = \text{MS}_B/\text{MS}_W$
- Post-hoc: Tukey HSD, Bonferroni
- Effect size η^2; two-way ANOVA

A/B Testing

- A/B test = randomized experiment
- MDE, sample size planning
- Pitfalls: peeking (>25% false pos.),
  too many metrics, novelty
- Sequential testing (O'Brien–Fleming)

Key lesson: Comparing k groups needs ANOVA, not $\binom{k}{2}$ t-tests. A/B testing = hypothesis testing for products.

Preview: Regression Inference & GLMs *(in ML course)*

L12 — Regression Inference

- $Y = X\beta + \varepsilon$, LINE assumptions
- SE of $\hat{\beta}_j$, t-tests, CIs
- Overall F-test, R^2, R^2_{adj}
- Gauss–Markov: OLS is BLUE
- Diagnostics: residuals, QQ, leverage, Cook's distance
- Logistic: Z-tests, odds ratios

L13 — GLMs

- Three components: Random + Systematic + Link
- Exponential family
- OLS = GLM(Normal, Identity)
- Logistic = GLM(Bernoulli, Logit)
- Poisson regression (log link)
- IRLS, deviance, AIC

Key lesson: OLS, logistic, and Poisson regression are all GLMs — same framework, different distribution and link function.

Preview: Causal Inference *(in ML course)*

The Problem

- Correlation $\neq$ causation
- Confounding, reverse causation, collider bias
- Potential outcomes: $Y(1), Y(0)$
- Fundamental problem: only one is observed
- ATE $= E[Y(1) - Y(0)]$

The Solutions

- RCTs / A/B tests (gold standard)
- DAGs: fork, chain, collider
- Backdoor criterion
- $\text{do}(X) \neq$ observe X
- Observational methods: regression, matching, propensity scores, IV

Key lesson: To claim causation, you need either randomization or causal assumptions (encoded in a DAG). Data alone is not enough.

The Formulas That Matter Most

MSE decomposition
$\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Var}$

MLE
$\hat{\theta}_{\mathrm{MLE}} = \arg\max_\theta \sum_i \log f(x_i \mid \theta)$

Confidence interval
$\hat{\theta} \pm z_{\alpha/2} \cdot \mathrm{SE}(\hat{\theta})$

GLM
$g(\mu) = X\beta, \;\; Y \sim \text{exp. family}$

Cramér–Rao bound
$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{n\mathcal{I}(\theta)}$$

MAP
$\hat{\theta}_{\mathrm{MAP}} = \arg\max_\theta \left[\ell(\theta) + \log \pi(\theta)\right]$

Likelihood ratio test
$$-2\log \frac{L(\hat{\theta}_0)}{L(\hat{\theta})} \mathrel{\dot{\sim}} \chi^2_k$$

ATE (causal)
$\mathrm{ATE} = E[Y(1)] - E[Y(0)]$

Five Themes That Connect Everything

1. Uncertainty is the point. Every estimate has a sampling distribution. Report SE, CI, p-value — not just the number.

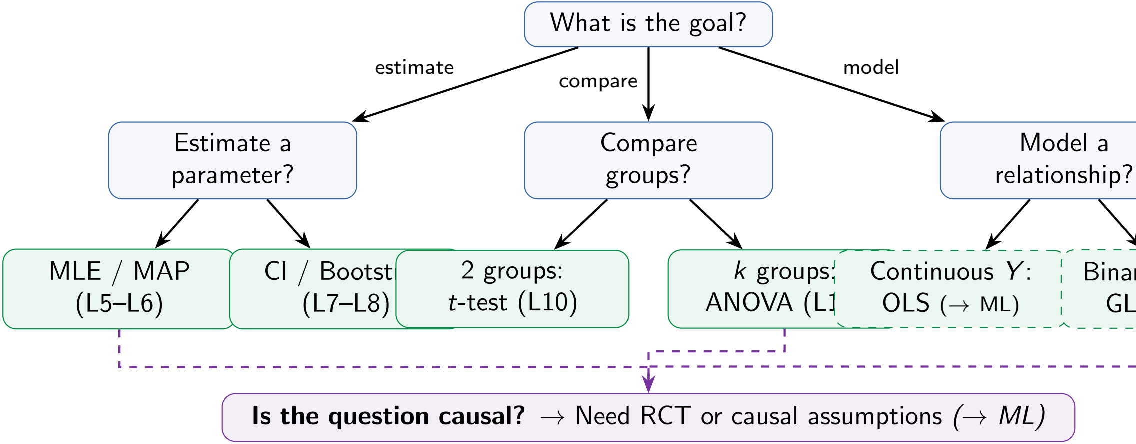
2. Bias-variance tradeoff is everywhere. MLE vs MAP. OLS vs ridge. Simple vs complex model. More data helps variance, not bias.

3. Likelihood is the universal language. MLE, LRT, Fisher info, deviance, AIC, GLMs — all built on $L(\theta \mid \text{data})$.

4. Assumptions matter more than methods. Every test and model has assumptions (normality, independence, linearity, no confounding). Violations matter.

5. Association $\neq$ causation. Regression finds patterns. Only experiments (or careful causal reasoning) find causes.

Which Method Do I Use?



Top Mistakes to Avoid

✗ Confusing SD and SE

SD $=$ spread of data.
SE $=$ spread of $\hat{\theta}$. SE $\to 0$ as $n \to \infty$.

✗ "$p > 0.05$ = no effect"

Absence of evidence $\neq$ evidence of absence. Could be low power.

✗ Ignoring assumptions

t-test on heavy-tailed data, OLS on binary Y, ANOVA with unequal variances.

✗ Peeking at p-values

Testing repeatedly as data arrives inflates false positives to $>25\%$.

✗ Correlation → causation

Regression finds association.
Causation needs a DAG or an experiment.

✗ R^2 worship

R^2 always increases with more predictors.
Use R^2_{adj} or AIC. Check diagnostics.

Your Python Toolkit

Task	Library	Key function
Descriptive stats	`numpy`, `pandas`	`.describe()`, `np.quantile()`
Plotting	`matplotlib`, `seaborn`	`sns.boxplot()`, `sns.histplot()`
MLE	`scipy.stats`	`.fit()`, `minimize(-loglik)`
Bootstrap	`scipy.stats`	`bootstrap()`
t-tests	`scipy.stats`	`ttest_ind()`, `ttest_rel()`
χ^2 tests	`scipy.stats`	`chi2_contingency()`
ANOVA	`scipy.stats`	`f_oneway()`
OLS regression *(ML)*	`statsmodels`	`OLS().fit().summary()`
GLMs *(ML)*	`statsmodels`	`GLM(family=...).fit()`
Logistic *(ML)*	`statsmodels`	`Logit().fit()`
Causal inference *(ML)*	`dowhy`	`CausalModel()`
Prop. scores *(ML)*	`causalml`	`PropensityScoreMatching()`

Where to Go Next

Machine Learning

Bias-variance, cross-validation, regularization, trees, neural networks

Bayesian Statistics

MCMC, Gibbs, hierarchical models, Bayesian regression, `PyMC`, `Stan`

Time Series

AR, MA, ARIMA, stationarity, forecasting, `statsmodels.tsa`

Causal ML

Heterogeneous treatment effects, causal forests, double ML

Experimental Design

Block designs, factorial experiments, multi-armed bandits, adaptive trials

That's Statistics!

10 lectures · From data to inference (and a glimpse beyond)

"All models are wrong, but some are useful."

— George E. P. Box

Bonus: The Delta Method *(slides we never got to in L8)*

The setup. You've estimated $\hat{\theta}$ with $\sqrt{n}\big(\hat{\theta} - \theta\big) \ \to \ N(0, \sigma^2)$.

The question. You actually care about a *transformed* quantity $g(\hat{\theta})$:

- $\hat{p} \ \to \$ log-odds $\log\big(\hat{p}/(1-\hat{p})\big)$
- $\bar{X}_n \ \to \$ rate $1/\bar{X}_n$ (events per unit time)
- $(\hat{\mu}_1, \hat{\mu}_2) \ \to \$ ratio $\hat{\mu}_1/\hat{\mu}_2$

What is the asymptotic distribution of $g(\hat{\theta})$?

Delta method. If g is differentiable at θ with $g'(\theta) \neq 0$,

$$\sqrt{n}\big(g(\hat{\theta}) - g(\theta)\big) \ \xrightarrow{d} \ N\big(0,\ [g'(\theta)]^2\sigma^2\big).$$

Idea: Taylor-expand g around θ, keep the linear term, transfer the variance.

Delta Method: CI on Log-Odds (Worked Example)

Problem. Observe $X \sim \mathrm{Bin}(n, p)$, $\hat{p} = X/n$. Build a 95% CI on the *log-odds* $\log\big(p/(1-p)\big)$.

Step 1. Sampling distribution of $\hat{p}$:

$$\hat{p} \mathrel{\dot\sim} N\left(p,\ \frac{p(1-p)}{n}\right)$$

Step 2. Compute $g'(p)$:

$$g(p) = \log\frac{p}{1-p}, \quad g'(p) = \frac{1}{p(1-p)}$$

Step 3. Apply the delta method:

$$\widehat{\log\text{-odds}} \mathrel{\dot\sim} N\left(\log\frac{p}{1-p},\ \frac{1}{n\,p(1-p)}\right)$$

Step 4. Plug in $\hat{p}$, build the CI:

$$\widehat{\mathrm{SE}} = \sqrt{\tfrac{1}{n\,\hat{p}\,(1-\hat{p})}}, \qquad \mathrm{CI} = \log\tfrac{\hat{p}}{1-\hat{p}} \pm 1.96\,\widehat{\mathrm{SE}}$$

Numbers. $X = 30$, $n = 100 \ \Rightarrow\ \hat{p} = 0.3$:

$$\widehat{\log\text{-odds}} = \log(0.3/0.7) \approx -0.847$$

$$\widehat{\mathrm{SE}} = \sqrt{\tfrac{1}{100\cdot 0.21}} \approx 0.218$$

$$\text{95\% CI:}\quad -0.847 \pm 0.428 = (-1.28,\ -0.42)$$

Why bother? Wald CI on log-odds, then back-transform → CI on odds, is more accurate than direct Wald CI on $\hat{p}$ when $\hat{p}$ is near 0 or